

APT-BENCH: PATIENT-DISJOINT HIERARCHICAL CELL TYPING AND MEASUREMENT-EFFICIENT DISEASE RECOGNITION FROM SINGLE-CELL APTAMER PROFILES

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ABSTRACT

Single-cell aptamer (APT) profiling offers a blood-based view of both disease-associated molecular state and circulating immune composition, but evaluating such data requires separating targets that live at different statistical units. Disease is a patient-level label, whereas lineage and subtype are cell-level labels arranged in a genuine hierarchy. We introduce APT-Bench, a patient-disjoint evaluation setting for multi-resolution characterization on a real-world APT liquid-biopsy cohort of 40 patients and more than 370,000 cells, spanning patient-level disease recognition, cell-level lineage and subtype annotation, protocol-sensitivity analysis, and measurement-efficiency analysis.

Across this benchmark, coarse lineage recognition is consistently easier than fine subtype recognition, revealing a granularity gap that is hidden by aggregate reporting. Encoder-matched controls show that **DropCASCADE** does not have a statistically supported architecture-specific advantage: its pooled coarse/fine macro-F1 is 0.326/0.152 versus 0.323/0.147 for the matched flat Transformer and 0.333/0.151 for its HCE-trained counterpart, with all paired patient-bootstrap differences spanning zero. Despite 361,792 evaluated cells, subtype performance remains abundance-sensitive: per-subtype F1 is more strongly associated with log cell count than with patient coverage across models, while most incorrect subtype predictions cross the true parent lineage. Patient-level disease recognition is extremely strong under the prespecified patient-disjoint folds, but stress tests show that this signal depends heavily on patient-specific absolute-expression structure. Finally, nested panel selection preserves the observed full-panel patient macro-F1 with 10 aptamers, and equal-cell subsampling preserves patient-level accuracy with as few as 10 cells per patient. These results position APT-Bench as a benchmark for patient-independent hierarchical cell typing, disciplined disease-level audit, and measurement-efficient APT analysis in small-cohort clinical machine learning.

1 INTRODUCTION

Cancer screening and diagnosis are usually framed at the patient level: does a person have disease, and if so which disease category is most likely? Single-cell liquid biopsy changes this framing. A single blood draw can support both a patient-level diagnostic decision and a cell-level characterization of the circulating immune compartment. These outputs are related, but they do not live at the same observational unit. Disease is a property of the patient, whereas lineage and subtype are properties of the cell.

This mismatch matters for evaluation. Our cohort contains more than 370,000 single cells, but only 40 patients. Treating cells as independent disease observations would substantially overstate the effective sample size of the diagnostic task. It would also hide a second asymmetry: lineage and subtype form a genuine cell-level hierarchy, but disease does not sit above that hierarchy. The clinically relevant problem is therefore not a single three-level taxonomy. It is a multi-resolution setting in which different targets require different evaluation protocols, inductive biases, and interpretations.

We study this setting on aptamer-based single-cell protein profiling. Aptamer (APT) measurements provide a fixed 293-dimensional panel of protein-target binding signals for each cell, offering a minimally invasive view of both disease-associated molecular state and immune composition from peripheral blood (Wu et al., 2025). This modality is attractive for repeated testing and potentially broad deployment, but it raises three intertwined modeling questions. First, how much predictive information is present at coarse versus fine biological resolution? Second, how sensitive is patient-level disease recognition to the mismatch between hundreds of thousands of correlated cells and only 40 independent patients? Third, how many aptamers and cells are needed to preserve disease recognition, and is that signal concentrated in a small unique panel or distributed across the assay?

We address these questions with APT-Bench, a patient-disjoint benchmark for multi-resolution characterization from single-cell APT profiles. The benchmark is organized around three primary tracks: cell-level coarse-to-fine hierarchy prediction, patient-level disease recognition with statistical-unit and confounding audits, and measurement-efficiency analysis over aptamer and cell budgets. This framing emphasizes that the central unresolved cell-level problem is not disease classification, which is already nearly saturated within cohort by simple baselines, but reliable hierarchical annotation under patient-independent evaluation.

To instantiate the hierarchy track, we use **DropCASCADE**, a differentiable lineage-subtype cascade. The model combines a shared APT encoder, soft lineage routing, and subtype experts. We compare it with classical models and encoder-matched flat and HCE-trained Transformer controls. It serves as a hierarchy-aware neural reference rather than a claimed universal winner; the matched controls test whether any gain survives control of encoder capacity and training budget.

This perspective leads to three empirical findings. First, coarse lineage recognition is substantially easier than fine subtype recognition across all models. A subtype-level audit shows that raw cell abundance explains more performance variation than patient coverage, but only partially, and that most wrong subtype predictions cross the true lineage rather than remaining among biological siblings. Second, disease labels are perfectly recoverable under the prespecified within-cohort patient folds, but sensitivity analyses show that this signal depends strongly on patient-specific absolute-expression structure, making overclaiming inappropriate in the absence of acquisition-batch metadata. Third, the disease signal is measurement-efficient but distributed: nested selection preserves patient-level performance with 10 aptamers, and equal-cell-per-patient subsampling preserves patient-level accuracy with as few as 10 cells per patient, while random-panel and remove-top controls argue against a unique indispensable biomarker set.

Our contributions are therefore as follows.

- **APT-Bench, a patient-disjoint benchmark for single-cell APT liquid biopsy.** We formalize coarse lineage prediction, fine subtype prediction, patient-level disease recognition, statistical-unit audits, and measurement-efficiency analysis within a single evaluation setting over 40 patients and more than 370,000 cells.
- **A subtype-centered account of the granularity gap.** Beyond aggregate macro-F1, we quantify how cell abundance and patient coverage relate to each subtype’s performance, decompose incorrect subtype predictions into sibling and cross-lineage errors, and report the coarse-to-fine gap for every model. We use **DropCASCADE** as a hierarchy-aware reference model rather than claiming universal superiority.
- **A disciplined disease-recognition and measurement-efficiency audit.** We show that disease recognition is extremely strong within cohort but sensitive to patient-specific absolute-expression structure, and that nested aptamer-panel selection plus equal-cell subsampling reveal a low measurement budget under the current within-cohort task.

2 RELATED WORK

Liquid biopsy and multi-cancer detection. Blood-based multi-cancer early detection has focused primarily on circulating tumor DNA, methylation, fragmentomics, and bulk protein assays (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). These approaches target a patient-level diagnostic decision. Our setting differs because APT profiling observes individual circulating cells, which introduces a second prediction problem at cell resolution: identifying

lineage and fine subtype structure within the same blood draw. The resulting task is not simply “more labels,” but a change in statistical unit and hierarchy structure.

Single-cell foundation models and modality mismatch. Large pretrained models such as Geneformer, scGPT, and scFoundation learn representations from transcriptomic data and are widely used as reference points for single-cell representation learning (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024). Their success does not transfer automatically to our setting. The input modality is different, since APT consists of a fixed 293-dimensional protein-oriented panel rather than a gene-vocabulary sequence. More importantly, recent evaluation work shows that frozen or zero-shot single-cell foundation-model embeddings are often matched by simpler baselines once the downstream task and evaluation protocol are made explicit (Kedzierska et al., 2025). This motivates training directly on APT profiles and treating evaluation design itself as a first-class contribution.

Hierarchical cell-type classification. Cell identity naturally admits coarse-to-fine structure, and recent work has explored losses or architectures that exploit this structure for annotation and out-of-distribution robustness (Jia et al., 2025; Cultrera di Montesano et al., 2026). Our use of hierarchy is narrower and more explicit: disease is kept outside the hierarchy, while lineage and subtype form the only genuine taxonomy. We compare **DropCASCADE** not only with flat classifiers but also with a matched flat FT-style Transformer control (Gorishniy et al., 2021) and the same control trained with a two-level adaptation of hierarchical cross-entropy (HCE), the most directly related hierarchy-aware loss.

Measurement-efficient disease recognition. Clinical translation of high-plex assays depends not only on accuracy but also on measurement cost. APT-Bench therefore includes a compact-panel analysis that asks how many aptamers are needed to preserve patient-level disease recognition under nested patient-disjoint evaluation. This is distinct from biomarker discovery: our goal is to quantify predictive budget and redundancy in the assay, not to claim that a small panel is a universal causal signature.

Tabular deep learning versus classical machine learning. APT expression is a medium-dimensional tabular signal. In that regime, gradient-boosted trees remain strong defaults and often rival or outperform deep tabular models (Grinsztajn et al., 2022; Erickson et al., 2026). This literature is directly reflected in our results: XGBoost and encoder-matched Transformer controls remain competitive at both resolutions, so **DropCASCADE** cannot be justified by a blanket accuracy claim. We instead use it as a structured reference for the hierarchy diagnostics, while disease recognition is audited with simple, strong tabular baselines.

3 PROBLEM SETUP

3.1 DATASET AND PREDICTION TARGETS

Our dataset contains single-cell aptamer (APT) protein expression profiles from peripheral blood collected from 40 patients across six disease groups: colorectal cancer (CRC), gastric cancer (GC), hepatocellular carcinoma (HCC), pancreatic cancer (PC), biliary tract cancer (BTC), and healthy controls (Normal), with 6–8 patients per group. The raw cohort contains 374,854 cells. After excluding the transcriptome-derived *Unknown* subtype label, which denotes uncertain annotation rather than a validated biological state, the main benchmark contains 361,792 cells. Each cell is represented by a 293-dimensional APT expression vector.

The 293 aptamers correspond to designed protein-target probes rather than an arbitrary learned basis. Expression values are log-transformed and standardized feature-wise using statistics estimated from the training partition only. Lineage and subtype labels are obtained from paired transcriptomic profiling on the same cells through barcode matching. The transcriptomic modality is used only to define supervision and is never provided as input to the models.

APT-Bench contains two kinds of targets at different observational units.

- **Patient-level disease recognition.** Each patient is assigned to one of six disease groups.

- **Cell-level hierarchy prediction.** Each cell is assigned to one of five coarse immune lineages and one of 27 fine subtypes. Every subtype maps to exactly one parent lineage through a manually curated lookup table.

Disease is therefore not treated as the root of a three-level hierarchy. It is a separate patient-level label observed on the same blood draw.

3.2 EVALUATION PROTOCOLS

We evaluate models under two main protocols with different purposes.

Patient-level 5-fold cross-validation. This is the primary protocol for all headline claims. Patients are partitioned into five disease-stratified folds, each containing eight held-out test patients. At every fold, preprocessing statistics, model fitting, and hyperparameter selection use only the training and validation patients from the remaining folds. We pool out-of-fold predictions across all five test folds, yielding a patient-disjoint evaluation spanning all 40 patients.

Cell-level split control. As a protocol-sensitivity control, cells are randomly split irrespective of patient identity. Cells from the same patient can therefore appear in both training and evaluation sets. We do not treat this as a realistic deployment protocol. Its only role is to quantify performance inflation caused by leakage.

3.3 CLASS DISTRIBUTION

Because the subtype distribution is long-tailed, we use macro-F1 as the primary metric for coarse-lineage and fine-subtype prediction. Accuracy is reported as a secondary metric. For uncertainty quantification under the main patient-level protocol, we compute 95% confidence intervals from 2,000 paired patient-level bootstrap resamples over the 40 pooled test patients.

From the same pooled predictions, we also report per-subtype support and F1, support–performance rank correlations, sibling versus cross-lineage errors, and each model’s coarse-minus-fine gap as descriptive difficulty diagnostics rather than new ranking metrics.

We report disease recognition at the patient level, since the clinical decision is made per patient rather than per cell. Cell-level disease accuracies are included only as supplementary evidence and are not interpreted as though hundreds of thousands of cells were independent disease observations.

4 HIERARCHY-AWARE REFERENCE MODEL

APT-Bench is intentionally broader than any single model. We nevertheless need a hierarchy-aware reference architecture that can operate directly on APT profiles under the benchmark protocols. **DropCASCADE** plays this role. It focuses on the genuine cell-level hierarchy, lineage $\rightarrow$ subtype, and leaves disease recognition to separate tabular baselines and patient-level aggregation. We do not present the architecture as a primary contribution: without full five-fold component ablations, its global path, coarse auxiliary objective, soft routing, and contrastive construction should be read as implementation choices rather than independently validated sources of performance.

4.1 CELL ENCODER

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i , we form a token

$$t_i = \text{Emb}(i) + \text{MLP}_{\text{val}}(x_i), \quad i = 1, \dots, 293. \quad (1)$$

A learnable [CLS] token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final [CLS] representation

$$z = \text{Encode}(x) \in \mathbb{R}^{256} \quad (2)$$

serves as the shared cell representation for downstream tasks.

4.2 SOFT-ROUTED HIERARCHICAL CASCADE

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \quad (3)$$

with routing distribution

$$p(m | x) = \text{softmax}(c/\tau)_m, \quad (4)$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, **DropCASCADE** uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha p(m(k) | x) e_{m(k),k}(z) + \gamma \log(\epsilon + (1 - \epsilon)p(m(k) | x)), \quad (5)$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m , $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg \max_k s_k(x), \quad \hat{m} = \arg \max_m c_m. \quad (6)$$

4.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\text{Proj}(z)}{\|\text{Proj}(z)\|_2} \quad (7)$$

be a normalized projection of the encoder representation. For anchor cell i , positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

4.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}} \mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t) \mathcal{L}_{\text{contrast}}, \quad (8)$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §4.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5} , gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling, as our experiments found that it did not improve performance after correcting an implementation issue; additional details are provided in Appendix A.8.

Matched Transformer controls. To isolate encoder capacity from hierarchical modeling, a flat FT-style Transformer control (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by **DropCASCADE**, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di

Aggregation / Representation	Patient Acc.	Patient Macro-F1	Macro-AUROC	95% CI (Acc.)
Cell-level classifiers*, majority vote	1.000	1.000	–	[0.912, 1.000]
Patient-level APT median + LR [†]	0.950	0.948	1.000	–
Gold subtype composition (CLR) + LR [‡]	0.575	0.572	0.842	–
Gold lineage composition (CLR) + LR [‡]	0.400	0.377	0.678	–

* All five cell-level classifiers (Logistic Regression, XGBoost, Linear SVM, Random Forest, Reference Correlation) reach the identical 40/40 majority-vote result and are therefore collapsed into a single row; see Table 4 for their (distinct) cell-level accuracies. [†] Genuinely patient-level feature (median APT expression per patient per aptamer, $n = 40$, no cell-level pseudo-replication). [‡] Gold-label immune-cell composition (paired transcriptomic annotation, not model-predicted subtypes); CLR = centered log-ratio transform of the compositional proportions. All rows use the same patient-level 5-fold partition (§3.2); the confusion matrix for the majority-vote rows is diagonal (40/40 patients correctly assigned; see Appendix A.1).

Table 1: **Patient-level** disease classification ($n = 40$ patients), the correct statistical unit for a per-patient diagnostic call. All five cell-level classifiers agree perfectly once cell-level predictions are aggregated to the patient by majority vote (95% Clopper-Pearson CI on the 40/40 result: [0.912, 1.000]). A model trained on purely patient-level summary features, with no cell-level pseudo-replication at all, achieves comparably strong discrimination, and immune-cell composition alone (gold-label, CLR-transformed) is also clearly above chance. Because the cohort contains only 40 patients (6–8 per disease) and no acquisition-batch metadata (plate, run, or date) exists for this dataset, we cannot exclude a disease-correlated technical contribution to this performance; see Appendix A.1 for the sensitivity analyses (patient-wise centering, patient-fingerprint diagnostic) that motivate this caveat, and §6 for a full discussion. This result should be read as within-cohort feasibility, not evidence of clinical generalization.

Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \text{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m . Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage.

Disease recognition is not handled by a dedicated head inside **DropCASCADE**. This is a deliberate modeling decision rather than a claim that disease is solved in a strong external sense: within-cohort disease prediction is already nearly saturated by simple classical baselines, while the architectural challenge addressed by **DropCASCADE** is the unresolved cell-level hierarchy problem.

5 RESULTS

5.1 DISEASE RECOGNITION IS STRONG WITHIN COHORT, BUT REQUIRES CAVEATS

We begin with disease because it is the clinically salient target, then immediately separate it from the central hierarchy benchmark analysis. Under the patient-level 5-fold protocol, all five classical cell-level baselines yield the same patient-level majority-vote result: all 40 patients are correctly assigned, for a Clopper-Pearson 95% interval of [0.912, 1.000] (Table 1). A genuinely patient-level model using each patient’s median APT profile also performs strongly, reaching accuracy 0.950 and macro-AUROC 1.000.

These results do not justify a claim of solved clinical diagnosis. Sensitivity analyses in Appendix A.1 show that the disease signal depends strongly on patient-specific absolute-expression structure: patient-wise centering reduces patient-level accuracy from 1.000 to 0.650, and a 40-way patient-identity diagnostic reaches 88.4% accuracy on held-out cells. Because acquisition-batch metadata are unavailable, the observed offsets cannot be cleanly decomposed into biological versus technical sources. We therefore use disease recognition primarily to motivate the need for careful statistical-unit accounting in APT-Bench, not as the main architectural success story.

Task	Model	Macro-F1	Accuracy	95% CI (Macro-F1)
Coarse lineage	DropCASCADE	0.326	0.502	[0.301, 0.352]
	Flat FT-style Transformer	0.323 ^{n.s.}	0.507	[0.302, 0.345]
	FT-style Transformer + HCE	0.333^{n.s.}	0.503	[0.312, 0.354]
	XGBoost	0.322 ^{n.s.}	0.517	[0.300, 0.345]
Fine subtype	DropCASCADE	0.152	0.305	[0.129, 0.166]
	Flat FT-style Transformer	0.147 ^{n.s.}	0.315	[0.121, 0.161]
	FT-style Transformer + HCE	0.151 ^{n.s.}	0.306	[0.128, 0.167]
	XGBoost	0.143 ^{n.s.}	0.323	[0.128, 0.158]

Table 2: Patient-disjoint 5-fold hierarchy results on pooled out-of-fold predictions from all 40 patients. The two FT-style Transformer controls match **DropCASCADE**’s APT tokenizer, encoder dimensions, optimizer, early stopping, training budget, and validation-based checkpoint rule. HCE is the two-level adaptation in §4.4. Intervals use 2,000 paired patient bootstrap resamples. Relative to **DropCASCADE**, [†] denotes a significantly lower baseline, [‡] a significantly higher baseline, and ^{n.s.} no significant difference.

5.2 PATIENT-DISJOINT CELL-LEVEL HIERARCHY PREDICTION REMAINS DIFFICULT

The harder unresolved problem is cell-level hierarchy prediction under patient-disjoint evaluation. Table 2 shows that coarse lineage recognition is consistently easier than fine subtype recognition. For **DropCASCADE**, coarse macro-F1 reaches 0.326 while fine macro-F1 is only 0.152; for the matched flat FT-style Transformer, the corresponding values are 0.323 and 0.147. This granularity gap appears across model families and indicates that APT captures broad immune identity more reliably than fine subtype structure.

The matched comparison changes the architectural conclusion. Relative to the flat Transformer, **DropCASCADE** differs by only +0.0036 coarse macro-F1 (paired patient-bootstrap 95% CI [−0.0078, 0.0146]) and +0.0055 fine macro-F1 (95% CI [−0.0021, 0.0125]). HCE has the highest coarse point estimate (0.333), but its difference from **DropCASCADE** is also unsupported: **DropCASCADE**–HCE is −0.0069 (95% CI [−0.0154, 0.0026]) at the coarse level and +0.0015 (95% CI [−0.0083, 0.0047]) at the fine level. Differences from XGBoost likewise do not exclude zero. Thus the experiment does not isolate a performance gain from the cascade: the defensible result is the persistent coarse-to-fine difficulty under patient-disjoint evaluation, not superiority of a routing mechanism.

A coarse–fine consistency diagnostic computed from these same pooled out-of-fold predictions is reported in Appendix A.6. We keep it as supporting evidence because consistency can be altered mechanically by constrained decoding and is not a standalone measure of predictive quality.

5.3 WHY 361,792 CELLS STILL YIELD ONLY 0.15 SUBTYPE MACRO-F1

Cell abundance explains more than patient coverage, but neither is sufficient. For subtype k , let n_k be its cell count, P_k its patient count, and $F_{1,k}$ its pooled out-of-fold F1. We compare $\rho_{\text{cell}} = \text{Spearman}(F_{1,k}, \log n_k)$ with $\rho_{\text{patient}} = \text{Spearman}(F_{1,k}, P_k)$. Table 3 shows that the former is larger for all eight models (0.486 versus 0.288 for **DropCASCADE**; 0.540 versus 0.275 for XGBoost). Patient coverage has limited resolution because 16 of 27 subtypes occur in all 40 patients. Abundance is still incomplete: tail subtype Plasma reaches $F1 \approx 0.61$, whereas head subtype Memory B-1 remains near 0.03. Appendix A.2 reports all requested per-subtype quantities.

Most subtype errors cross lineage boundaries. We define $R_{\text{sibling}} = \Pr(m(\hat{k}) = m(k) \mid \hat{k} \neq k)$ using the fixed parent map $m(\cdot)$. Only 25.0% of **DropCASCADE** errors and 24.6% of XGBoost errors are sibling confusions; across all models, 70.5–80.9% of errors cross the true lineage (Table 3). Low fine-level F1 is therefore not primarily benign confusion among nearby siblings.

The granularity gap is a benchmark finding. For each model, $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$ is positive and ranges from 0.099 to 0.200. We use G only to expose persistent coarse-to-fine difficulty, not as a new metric contribution.

Model	ρ_{cell}	ρ_{patient}	G	R_{sibling}	Cross-lineage
DropCASCADe	0.486	0.288	0.174	0.250	0.750
Flat FT-style Transformer	0.505	0.255	0.176	0.243	0.757
FT-style Transformer + HCE	0.513	0.318	0.183	0.257	0.743
XGBoost	0.540	0.275	0.179	0.246	0.754
Logistic Regression	0.435	0.107	0.167	0.295	0.705
Linear SVM	0.339	-0.012	0.200	0.265	0.735
Random Forest	0.661	0.418	0.158	0.191	0.809
Reference Correlation	0.376	0.003	0.099	0.225	0.775

Table 3: Subtype support, granularity, and error diagnostics on pooled patient-disjoint predictions. The two ρ columns are Spearman associations of per-subtype F1 with log cell count and patient coverage. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among incorrect subtype predictions, R_{sibling} retains the true parent lineage and Cross-lineage is its complement. These are descriptive benchmark findings, not proposed ranking metrics.

5.4 PROTOCOL SENSITIVITY: PATIENT-LEVEL VERSUS CELL-LEVEL SPLITTING

The cell-level split control (Appendix A.3) makes both DropCASCADe and XGBoost look stronger, but the inflation is larger for DropCASCADe: +0.090 versus +0.050 on coarse lineage, and +0.108 versus +0.069 on fine subtype. Leakage can therefore change not only absolute performance but also the apparent advantage of a method family. APT-Bench treats patient-disjoint evaluation as a non-negotiable requirement, not a reporting detail.

5.5 DISEASE SIGNAL IS MEASUREMENT-EFFICIENT BUT PREDICTIVELY DISTRIBUTED

The compact-panel analysis in Appendix A.7 provides a final perspective on what the APT modality supports. Under a nested patient-level selection pipeline, all five outer folds choose the same development tolerance budget, $B^* = 10$, and the resulting 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on every outer test fold. A complementary equal-cell-per-patient subsampling diagnostic, computed from held-out disease predictions without retraining, preserves 100% patient-level accuracy across 50 random trials with only 10 cells per patient. At the same time, random panels approach full performance as the aptamer budget increases, and removing many top-ranked aptamers often leaves disease performance nearly unchanged. The correct interpretation is therefore not that the dataset isolates a unique minimal biomarker set, but that within-cohort disease information is highly redundant and can be recovered with substantially reduced aptamer and cell-count budgets.

Compact-panel recovery does not resolve whether the disease signal is biological, technical, or mixed. It instead shows a central asymmetry: within-cohort disease recognition is statistically and operationally easier than patient-disjoint cell-level hierarchy prediction.

6 DISCUSSION

APT-Bench shows that multi-cancer liquid biopsy decomposes into targets with different statistical units and evidentiary standards. Across matched models, coarse lineage prediction is easier than fine subtype prediction; abundance explains more subtype variation than patient coverage, yet most errors still cross lineage boundaries. Disease recognition is strong within cohort but vulnerable to patient-specific absolute-expression effects. Compact aptamer and cell budgets preserve this signal, indicating measurement efficiency rather than unique causal biomarkers.

Matched controls show no significant DropCASCADe advantage at either resolution; persistent cross-lineage errors instead point toward better patient-independent representations.

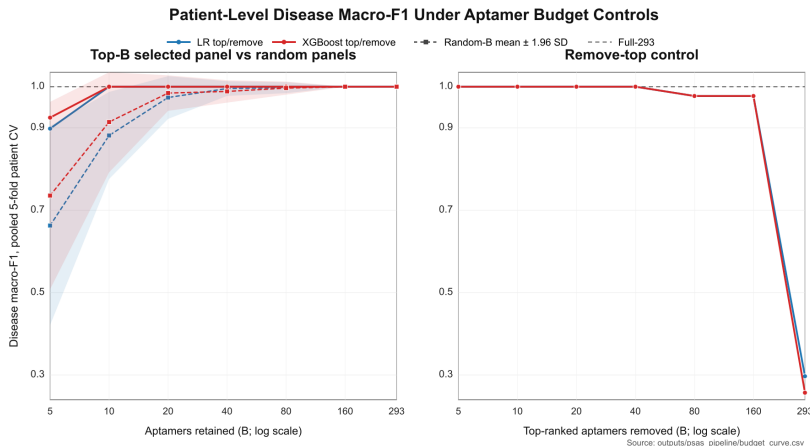


Figure 1: Patient-level disease macro-F1 under aptamer-budget controls. Left: Top- B and Random- B retained panels (mean ± 1.96 SD for random panels). Right: Remove-Top- B , with a separate x-axis denoting aptamers removed. The dashed line is the full 293-aptamer reference.

6.1 LIMITATIONS

Disease-level novelty is exploratory. The appendix leave-one-disease-out stress test has too few disease groups and too little proxy-novel diversity to support a main claim.

The disease result is within-cohort and potentially confounded. Only 40 patients are available and acquisition-batch metadata are missing; patient fingerprints and centering sensitivity show that biological and technical offsets cannot be separated.

Matched baselines narrow the method claim. The flat and HCE controls prevent attributing Transformer capacity to the cascade. They do not support a DropCascade-specific gain, so it remains a reference implementation rather than a superior model.

Compact panels imply measurement efficiency, not unique biomarkers. Small-panel and equal-cell analyses show distributed predictive signal and lower measurement budgets, not a universal causal biomarker set.

Future work requires external cohorts with batch metadata and out-of-cohort validation of subtype models and compact panels.

7 CONCLUSION

APT-Bench separates patient-disjoint hierarchy typing, patient-level disease audit, and measurement efficiency. Disease recognition remains strong under compact budgets but is sensitive to patient-specific structure, while subtype typing remains difficult despite 361,792 cells. Matched Transformer and HCE controls do not support a DropCASCADe-specific gain; the persistent granularity gap and cross-lineage errors instead motivate more patient-independent representations.

AI USE STATEMENT

ETHICS STATEMENT

REPRODUCIBILITY STATEMENT

AUTHOR CONTRIBUTIONS

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A APPENDIX

A.1 DISEASE CLASSIFICATION: CELL-LEVEL SOURCE PREDICTION AND PATIENT-LEVEL SENSITIVITY ANALYSES

Cell-level prediction of the disease source of a held-out patient. Table 1 reports patient-level disease classification, the correct statistical unit for a per-patient diagnostic call. For reference and comparability with earlier reporting conventions, Table 4 additionally reports the pooled *cell-level* metric, in which every held-out cell is treated as an independent prediction target and pooled across all 361,792 cells from the five test folds. This is an easier and statistically different task than patient-level diagnosis, since cells from the same patient are correlated observations rather than independent draws and patients with more cells contribute more weight. We therefore report this metric only as a lower-level diagnostic.

Model	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.
Logistic Regression	0.991	0.989	0.991	0.989
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation [†]	0.921	0.909	0.921	0.915

[†] SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 4: **Cell-level** prediction of the disease source of a held-out patient’s cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the patient-level diagnosis reported in Table 1: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient’s absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient’s own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 0.989 $\rightarrow$ 0.348), and patient-level majority-vote accuracy drops from 40/40 to

26/40. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient’s median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch meta-data.

A.2 PER-SUBTYPE SUPPORT AND PERFORMANCE AUDIT

Table 5 gives the complete subtype-level quantities underlying the coverage analysis in §5.3. Counts and patient coverage are computed from the pooled ground-truth labels, while F1 is computed from the same patient-disjoint out-of-fold predictions used for the main results. Mean cells per patient averages only over patients in which the subtype is observed. The table includes both **DropCASCADE** and XGBoost to show that the main abundance-versus-coverage pattern is not specific to one architecture.

Table 5: Per-subtype support and pooled out-of-fold F1. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	Lineage	Cells	Patients	Cells/patient	Regime	DropCascade F1	XGB F1
Erythrocyte	Other	65	6	10.8	Tail	0.253	0.133
Atypical Memory B	B	781	34	23.0	Tail	0.005	0.005
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027	0.014
platelet	Other	1,574	38	41.4	Tail	0.078	0.063
Tcm	T	2,121	36	58.9	Tail	0.000	0.004
Memory B-2	B	2,130	33	64.5	Tail	0.017	0.011
Plasma	B	2,173	40	54.3	Tail	0.611	0.599
pDC	Myeloid	2,615	40	65.4	Mid	0.029	0.019
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258	0.177
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020	0.023
GCB	B	3,283	31	105.9	Mid	0.039	0.047
Cycling T	T	3,667	40	91.7	Mid	0.028	0.012
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075	0.078
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069	0.078
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111	0.117
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009	0.002
cDC2	Myeloid	7,376	40	184.4	Mid	0.132	0.125
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019	0.017
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152	0.118
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188	0.188
Naive T	T	17,960	40	449.0	Head	0.131	0.149
Memory B-1	B	23,722	40	593.0	Head	0.034	0.032
CD8 Tem-1	T	26,714	40	667.9	Head	0.187	0.162
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437	0.451
CD56 NK	NK	38,442	40	961.0	Head	0.398	0.437
CD16 NK	NK	58,507	40	1462.7	Head	0.383	0.376
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417	0.435

For continuity with common long-tail reporting, Table 6 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	Tail Macro-F1	Head Macro-F1
DropCASCADe	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 6: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 5.

A.3 PROTOCOL SENSITIVITY: PATIENT-LEVEL VERSUS CELL-LEVEL SPLITTING

Table 7 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	DropCASCADe	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	DropCASCADe	0.152	0.260	+0.108

Table 7: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for **DropCASCADe** than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

A.4 LINEAGE-RESOLVED DISEASE-SPECIFIC APTAMER SIGNATURES

To assess whether the APT panel captures molecular differences beyond cell-type composition, we examined, within each of the three major immune lineages (T/NK, B, and myeloid), the aptamer whose average expression was most enriched in a given disease relative to the other five groups (disease-specific aptamer, selected independently per lineage). Figure 2 shows the resulting six disease-specific aptamers (one per disease category, including Normal) within each lineage. Several aptamers show a consistent, lineage-independent pattern: for example, the HCC-specific aptamer is elevated in HCC relative to all other categories in the T/NK, B, and myeloid compartments alike, suggesting a disease-associated signal that is not restricted to a single cell type. This lineage-resolved characterization is descriptive and based on the full panel prior to any train/test partitioning; it is intended to establish molecular plausibility for the APT modality rather than as a validated biomarker panel, and none of these aptamers are used as hand-selected features by **DropCASCADe** or any baseline in §5, which instead consume the full 293-dimensional panel.

A.5 A EUCLIDEAN HEURISTIC INSPIRED BY HYPERBOLIC HIERARCHY EMBEDDING

Motivated by the observation that tree-structured label hierarchies can be embedded with lower distortion under hyperbolic geometry than Euclidean geometry (Nickel & Kiela, 2017), we explored a lightweight Euclidean heuristic that directly enforces hierarchy-consistent distances between subtype representations, without the optimization overhead of a full hyperbolic parameterization (e.g., Riemannian optimization over a Poincaré ball). For each training batch we compute the

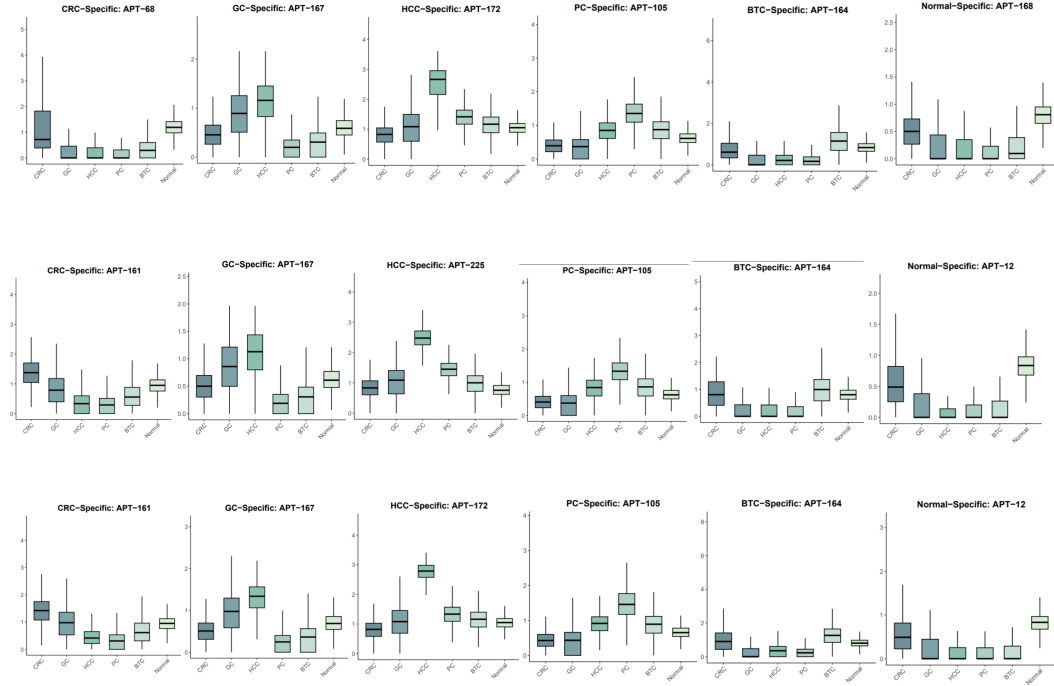


Figure 2: Disease-specific aptamer expression (mean per-cell value, boxplots over cells) within the T/NK (top), B (middle), and myeloid (bottom) lineages. Each panel shows the aptamer most enriched in the named disease relative to the other five categories, computed independently within that lineage.

mean encoder representation (§4.1) of every subtype present in the batch as a prototype, and add a margin-based regularizer that penalizes same-lineage subtype prototypes farther apart than a margin $m_{\text{intra}} = 1.0$ and different-lineage subtype prototypes closer together than a margin $m_{\text{inter}} = 2.0$, added to the training objective (§4.4) with weight 0.1.

Under the single patient-independent split used for architecture development (§3.2), this regularizer did not improve performance and in fact reduced it relative to the same configuration without the term: fine-subtype macro-F1 decreased from 0.171 to 0.136, and coarse-lineage macro-F1 decreased from 0.360 to 0.317. We did not conduct a hyperparameter sweep over the margin values or loss weight, so this diagnostic cannot determine whether the result reflects redundancy, optimization conflict, or an unsuitable configuration. We report the negative single-split result rather than omitting it, but do not treat it as a component ablation or evidence for the necessity of any retained **DropCASCADE** component.

A.6 COARSE-FINE CONSISTENCY DIAGNOSTIC

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 8 reports their natural agreement. Unlike an earlier single-split analysis, this version uses the same five-fold pooled out-of-fold prediction artifacts as the main hierarchy results, so every patient contributes exactly once to both evaluations.

Outer Fold	δ	B^*	LR Macro-F1	XGBoost Macro-F1
0	0.02	10	1.000	1.000
1	0.02	10	1.000	1.000
2	0.02	10	1.000	1.000
3	0.02	10	1.000	1.000
4	0.02	10	1.000	1.000
All folds	0.02	10	1.000	1.000

Table 9: Nested patient-level panel selection for disease recognition. In each outer fold, aptamer ranking and budget selection are performed using outer-training patients only, and the selected budget is evaluated once on the untouched outer-test patients. Under the prespecified tolerance margin $\delta = 0.02$, all five folds select the same development budget $B^* = 10$, and the resulting 10-aptamer panels preserve the observed full-panel patient macro-F1 for both Logistic Regression and XGBoost. This result supports a measurement-efficiency interpretation rather than a claim of unique causal biomarkers.

Model	Coarse-Fine Consistency
DropCASCAD E	0.812
Flat FT-style Transformer	0.795
FT-style Transformer + HCE	0.834
XGBoost	0.736
Logistic Regression	0.568
Linear SVM	0.631
Random Forest	0.573
Reference Correlation	0.380

Table 8: Natural coarse-fine consistency on the same 361,792-cell, 40-patient pooled out-of-fold artifacts as Table 2. For each model, the hard fine prediction is mapped to its parent and compared with the corresponding hard coarse prediction. *Unknown* is excluded. Consistency is supporting evidence rather than predictive superiority because it can be changed by constrained decoding.

A.7 MINIMAL APTAMER PANEL FOR DISEASE CLASSIFICATION

Because a smaller aptamer panel would be cheaper and simpler to deploy clinically, we asked how many of the 293 aptamers are actually needed to preserve the observed within-cohort disease result. We therefore performed a nested patient-level feature-selection analysis in which aptamer ranking and budget selection are carried out using outer-training patients only, while outer-test patients are never used for ranking or for choosing the final budget. Within this protocol, all five outer folds select the same development tolerance budget, $B^* = 10$, under a prespecified $\delta = 0.02$ criterion relative to the full panel.

The resulting development-selected 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on all five outer test folds. The broader budget curve in Figure 1 shows that top-ranked panels improve rapidly with panel size, but random panels also approach full performance as the number of retained aptamers increases, and removing the top-ranked aptamers often leaves performance nearly unchanged. A separate equal-cell-per-patient subsampling diagnostic preserves 100% patient-level accuracy across 50 random trials with only 10 cells per patient. We therefore interpret these results as evidence of measurement efficiency rather than as proof that disease information is concentrated in a unique indispensable biomarker set.

The fold-overlap ranking is also stable. The highest-ranked aptamers by mean outer-fold rank are APT-179, APT-225, APT-164, APT-166, APT-57, APT-69, APT-125, APT-161, APT-172, and APT-218, and 290 of the 293 aptamers are retained after a simple redundancy screen at $|r| \leq 0.9$. This again suggests that predictive signal is distributed across the panel. The descriptive lineage-resolved marker analysis in Appendix A.4 remains useful here: several aptamers highlighted independently

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	0.075	0.272	0.277	0.470	0.070

Table 10: Leave-one-disease-out generalization for **DropCASCADE** over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

by that descriptive analysis also appear near the top of the discriminative ranking, providing qualitative cross-support without turning either analysis into a causal biomarker claim.

A.8 BALANCED SAMPLING ABLATION

The final **DropCASCADE** configuration (§4.4) does not use class-balanced sampling. We initially implemented a balanced subtype-disease batch sampler to counteract the long-tailed subtype distribution (§3.3), but found and fixed an implementation issue in which `steps_per_epoch` was computed from the natural data loader’s batch size (512) rather than the sampler’s own, much smaller, per-batch cell count (e.g., 14 subtypes $\times$ 2 diseases $\times$ 4 cells = 112); this mismatch caused each nominal epoch to cover only about 22% of the training set when the sampler was enabled, silently under-training the model and desynchronizing every epoch-indexed schedule (warmup, contrastive ramp-up, routing ramp-up, early-stopping patience) relative to the sampler-disabled baseline.

After fixing this issue, the balanced sampler improved fine-subtype macro-F1 from 0.102 to 0.119 on the single patient-independent split used for architecture development, but both remain below the 0.171 achieved without any balanced sampling at all. We report this ablation as a diagnostic finding about our specific sampler implementation, not as evidence against balanced sampling in general, and did not further redesign the sampler given this result; the final configuration therefore omits it.

A.9 EXPLORATORY DISEASE-LEVEL NOVELTY TRACK

We treat disease-level novelty as an exploratory appendix track rather than a main contribution. In this protocol, all patients from one disease category are withheld during training and evaluated as a disease-level distribution shift. This is useful for stress-testing representations, but it is not a fair basis for method ranking because the cohort contains only six disease groups, proxy-disease supervision is limited, and the supervised detector and training-free confidence scores do not receive matched novelty supervision.

Table 11 reports two standard training-free scores, maximum softmax probability (Hendrycks & Gimpel, 2016) and energy (Liu et al., 2020), alongside a proxy-supervised novelty head. For each held-out disease, the cascade is trained only on the known diseases. The encoder is then frozen, and a lightweight MLP novelty head is trained to distinguish one proxy-novel known disease from the remaining known diseases. The truly held-out disease is never used during either cascade training or novelty-head training.

Under this exploratory nested leave-one-disease-out protocol, the proxy-supervised head attains higher descriptive AUROC than the training-free confidence scores. This should be interpreted narrowly: proxy disease supervision can help separate a withheld disease from known diseases in this cohort. It does not establish that the novelty head is intrinsically better than energy scoring, because the comparison lacks matched supervised probes on raw APT features, classical classifiers,

Held-out Disease	Proxy-Sup. Head AUROC	Proxy-Sup. Head AUPR	Energy AUROC
BTC	0.782	0.855	0.591
CRC	0.874	0.901	0.779
GC	0.729	0.698	0.575
HCC	0.881	0.842	0.558
Normal	0.607	0.843	0.722
PC	0.999	0.999	0.539
Macro-average	0.812	0.856	0.627

Table 11: Exploratory novelty detection under the nested leave-one-disease-out protocol. For each held-out disease, the proxy-supervised head is learned on a proxy-novel disease drawn from the remaining training diseases and then evaluated on the truly held-out disease, which is unseen during both cascade and novelty-head training. Energy is a training-free confidence score and does not receive matched proxy-novel supervision, so this table is descriptive rather than a fair method comparison. AUPR additionally depends on the held-out-disease prevalence in each evaluation set.

patient-clustered uncertainty intervals, proxy-choice sensitivity, and a formal ID/OOD patient manifest.